

Prem Rajesh Sargara

+91 94283 67330 | sargarapremrajesh@gmail.com | linkedin.com/in/gecdhd-comp-prem-sargara | github.com/PREMRAJESH

SUMMARY

Software Developer specializing in AI, Machine Learning, and Full-Stack engineering. Experience building AI-powered applications, machine learning projects, and full-stack software solutions. Seeking an AI/Machine Learning Engineering internship to scale intelligent systems and contribute to applied AI projects.

EDUCATION

Government Engineering College

Bachelor of Engineering in Computer Engineering (5th Semester)

Dahod, Gujarat

Aug 2023 – Present

Government Polytechnic College

Diploma in Computer Engineering | GPA: 8.33/10

Dahod, Gujarat

Sept 2020 – June 2023

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

AI/ML & Deep Learning: TensorFlow, Keras, Scikit-learn, OpenCV, Grad-CAM, NumPy, Pandas, Matplotlib

LLM & Generative AI: Google Gemini API, RAG Pipelines, LangFlow, Prompt Engineering

Web & App Development: React.js, React Native, Node.js, Flask, Tailwind CSS, Redux Toolkit

Databases & DevOps: MySQL, MongoDB, Firebase, Supabase, Cassandra, Git, GitHub Actions

PROJECTS

NeuroScan: AI Brain Tumor Detection | *Python, TensorFlow, Flask* | [GitHub](#)

- Built and deployed an end-to-end web application leveraging a fine-tuned VGG-16 Convolutional Neural Network (CNN) for MRI-based brain tumor classification.
- Trained and evaluated the model on a dataset of 3,264 MRI scans across 4 tumor classes, achieving a 90.4% validation accuracy and an F1 score of 0.89.
- Integrated Grad-CAM explainability to generate heatmaps for tumor regions, improving model interpretability, and implemented a Flask REST API for real-time inference.

StudyFlow: LLM Productivity Platform | *TypeScript, React, Gemini API* | [GitHub](#)

- Developed an intelligent study planner utilizing prompt engineering and the Google Gemini API to dynamically generate personalized, adaptive study schedules.
- Engineered automated PDF syllabus parsing and context-aware task recommendations, streamlining goal-setting workflows for users.
- Integrated real-time progress monitoring and a Pomodoro tracking system with a responsive frontend built on React and Tailwind CSS.

RAG-Based Knowledge Assistant | *LangFlow, LLMs, Semantic Search* | [GitHub](#)

- Designed a Retrieval-Augmented Generation (RAG) pipeline to deliver context-aware, grounded QA capabilities over custom document bases.
- Implemented semantic search workflows and optimized vector embedding strategies, improving answer relevance using retrieval-augmented document search.

Debate Verse AI | *TypeScript, LLM APIs* | [GitHub](#)

- Built an AI debate platform to conduct real-time, automated logical debates using LLM APIs.
- Implemented multiple Gemini-powered AI roles (Pro, Opponent, Fact Checker, Judge) for debate generation, fact-verification, and evaluation.

NimbusX: Real-Time Mobile Messaging | *React Native, Supabase, Redux* | [GitHub](#)

- Built a scalable, cross-platform mobile messaging application supporting real-time synchronization, private/group chats, and media sharing.

EXPERIENCE

Web Development Intern

Sept 2022 – Nov 2022

Brainy Beam Technologies Pvt. Ltd.

Ahmedabad, Gujarat

- Developed and deployed dynamic UI components for live client web applications, collaborating within an Agile engineering team.
- Optimized application performance and resolved critical frontend bugs under the mentorship of senior developers, ensuring scalable codebase architecture.
- Utilized Git for version control to maintain seamless continuous integration workflows and code repository hygiene.